



Blockchain Principles and Applications

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Recap

Proof of Stake

Allow meaningful participation based on stake
block proposers own coins

Level of participation proportional to stake
higher probability of being a proposer

Doing work is replaced by owning coins
energy efficient
capital efficient – no need for mining hardware

Idea 1

PoS attempt 1

$H(\text{hash}(\text{parent.header}), \text{Merkle root of tx}, \text{public key}) < \text{threshold} \times \text{stake}$

Idea 2

PoS attempt 2

$H(\text{hash}(\text{parent.header}), \text{public key}) < \text{threshold} \times \text{stake}$

Got rid of transaction hash

Problem: Liveness

just one trial to form a block

unlike PoW where nonce can be tried at will

Idea 3

PoS attempt 3

$H(\text{hash}(\text{parent.header}), \text{timestamp}, \text{public key}) < \text{threshold} \times \text{stake}$

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PoS attempt 3

$H(\text{hash}(\text{parent.header}), \text{timestamp}, \text{public key}) < \text{threshold} \times \text{stake}$

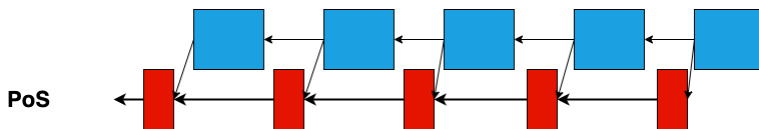
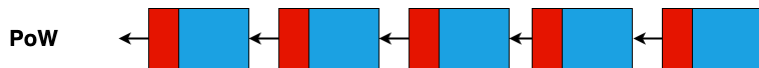
Problems:

1. Block content is not tamper-resistant against an adaptive adversary
2. Public election: vulnerable to bribery/corruption, not resistant to adaptive adversary

Idea 3

PoS attempt 3

$H(\text{hash}(\text{parent.header}), \text{timestamp}, \text{public key}) < \text{threshold} \times \text{stake}$



Block header



Block content

Idea 4

To solve the predictability of the winner in the future:

PoS attempt 4

$\text{hash}(\text{hash}(\text{parent.header}), \text{timestamp}, \text{secret key}) < \text{threshold} \times \text{stake}$

What is the problem?

Verifiable Random Function (VRF)

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$$\text{VRF}(\text{sk} , x) \rightarrow (y , \pi)$$

$$\text{Verify}(\text{pk} , x , y , \pi) \rightarrow \text{True/False}$$

- used for signing blocks
- helps achieve **adaptive security**

attacker corrupts upcoming blocks well in advance

Idea 4

PoS attempt 4

$\text{VRF}(\text{hash}(\text{parent.header}), \text{timestamp}, \text{secret key}) < \text{threshold} \times \text{stake}$

Attack: Nothing at Stake

$\text{VRF}(\text{hash}(\text{parent.header}), \text{timestamp}, \text{secret key}) < \text{threshold} \times \text{stake}$

Longest chain rule

parent is tip of longest chain

Adversary deviates

can grow on all blocks (even Genesis)

No computation limit to deviation

unlike PoW

Nothing at Stake (NaS)

NaS Tree

Honest participants

grow chain as a Poisson process

growth rate linear in time

Adversary

grows a tree of blocks

number of blocks grows exponentially in time

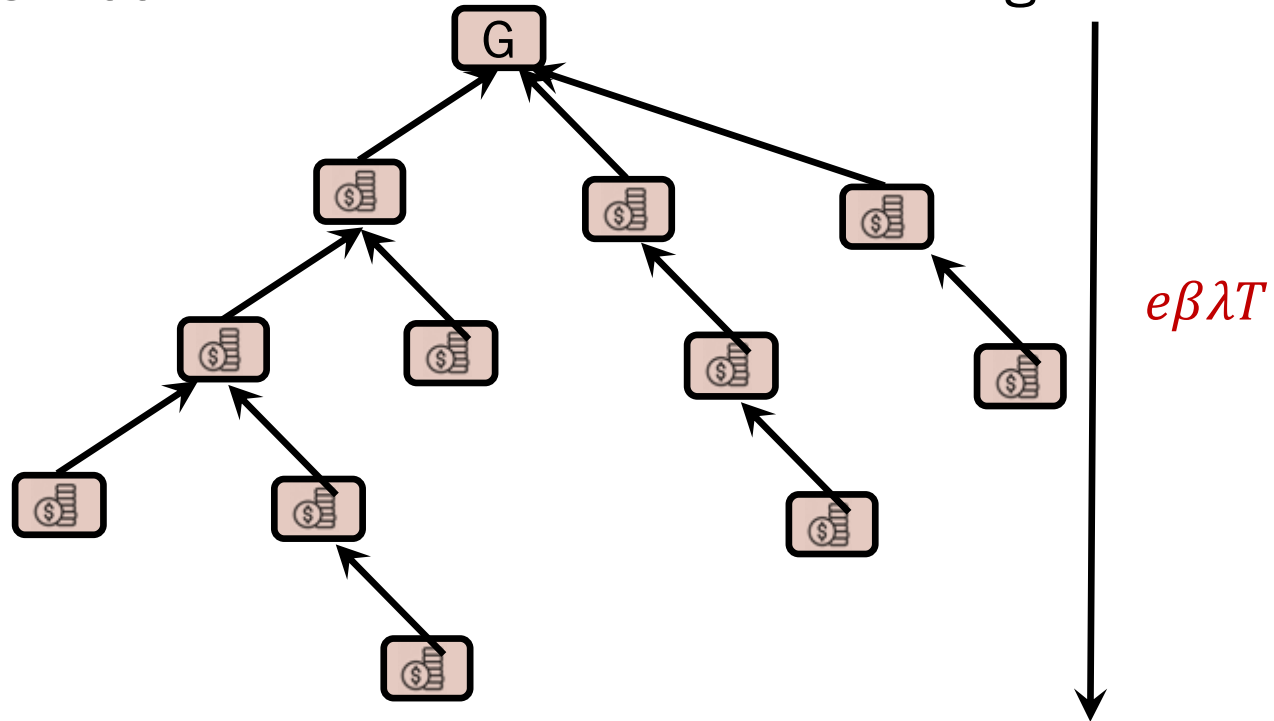
NaS

allows adversary to compete with honest participants unevenly

NaS Tree and Longest Chain

NaS Tree

Longest chain



Security of PoS Longest Chain

Honest participants

grow chain as a Poisson process

growth rate linear in time $(1 - \beta)\lambda T$

Adversary

grows a NaS tree in private

longest chain length $e\beta\lambda T$

Security against Private attack

$$(1 - \beta)\lambda T > e\beta\lambda T \quad \text{or} \quad \beta < \frac{1}{1+e} \approx 27\%$$

Boosting Security Threshold

Security against all attacks

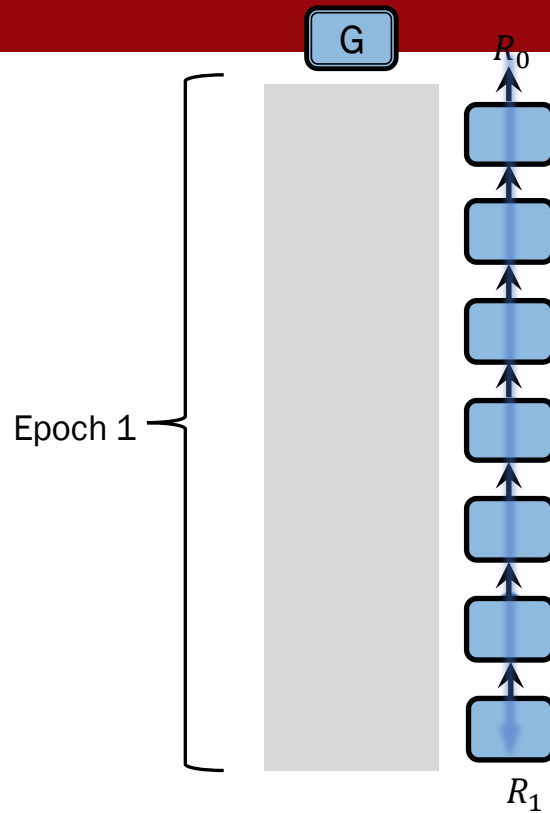
$$\beta < \frac{1}{1+e} \approx 27\%$$

Key idea: Reduce the number of randomness sources

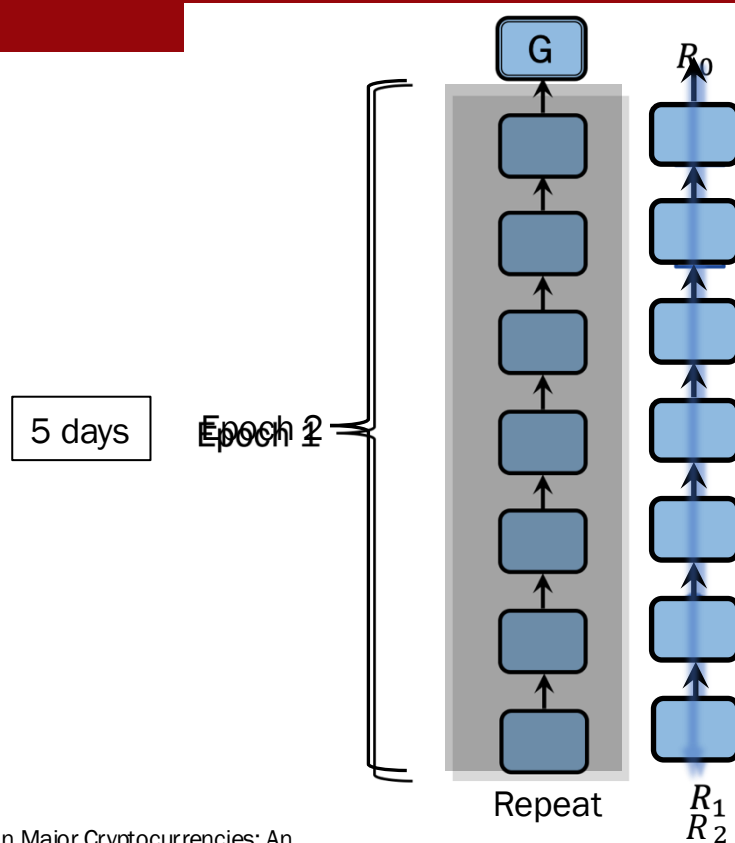
Idea 1: Only use randomness from genesis block

VRF(hash(Genesis), timestamp, secret key) < threshold x stake

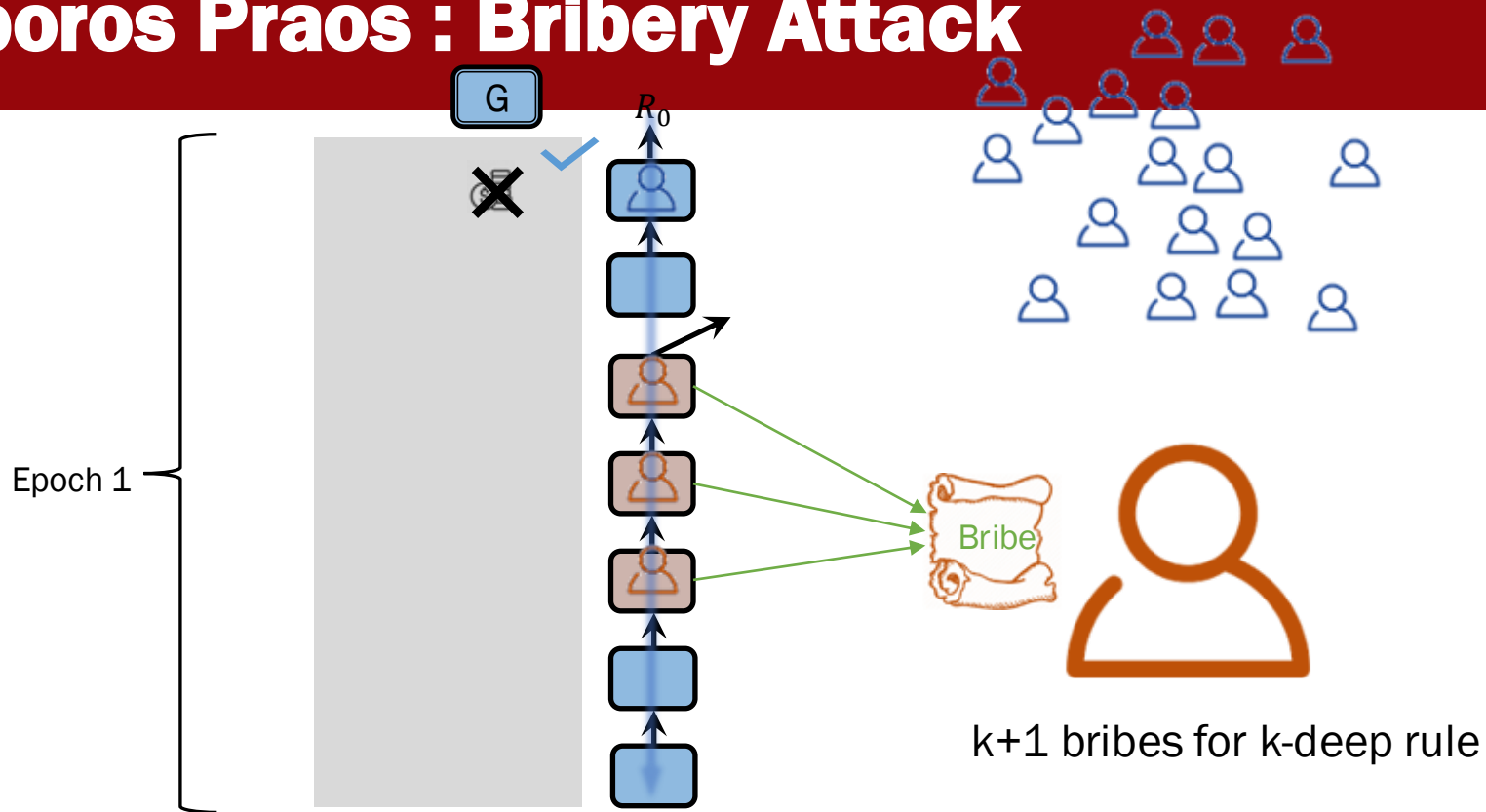
Ouroboros Praos



Ouroboros Praos



Ouroboros Praos : Bribery Attack



Boosting Security Threshold

Security against all attacks

$$\beta < \frac{1}{1+e} \approx 27\%$$

Key idea: Reduce the number of randomness sources

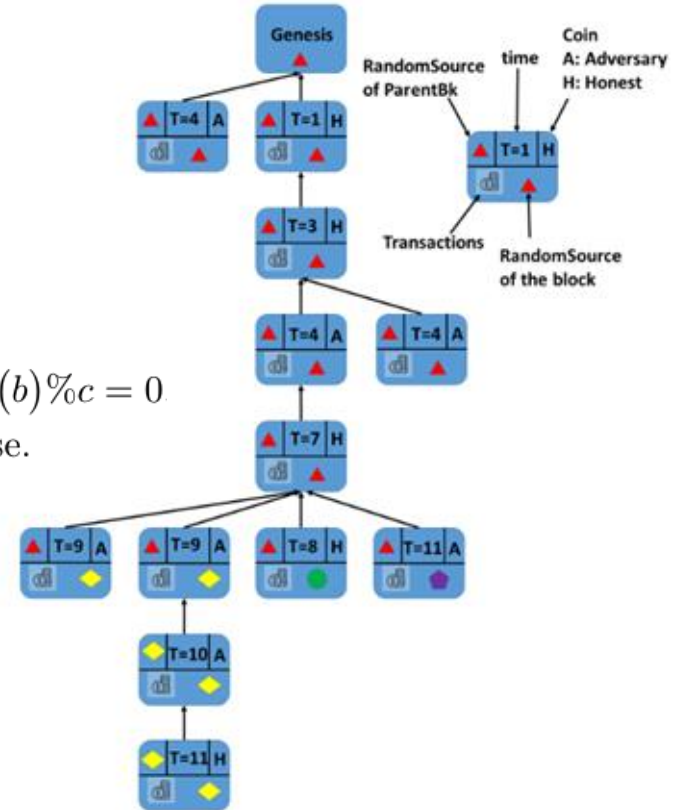
Idea 2: c-correlation

c-correlation

- Update the randomness of a block only at blocks with height multiples of c

$$\text{RandSource}(b) := \begin{cases} \text{VRF}(\text{RandSource}(\text{parent}(b)), ts, sk), & \text{if } \text{depth}(b) \% c = 0 \\ \text{RandSource}(\text{parent}(b)), & \text{otherwise.} \end{cases}$$

$$\text{VRF}(\text{RandSource}(\text{parent}), ts, sk_n) < T \cdot \text{stake}_n$$

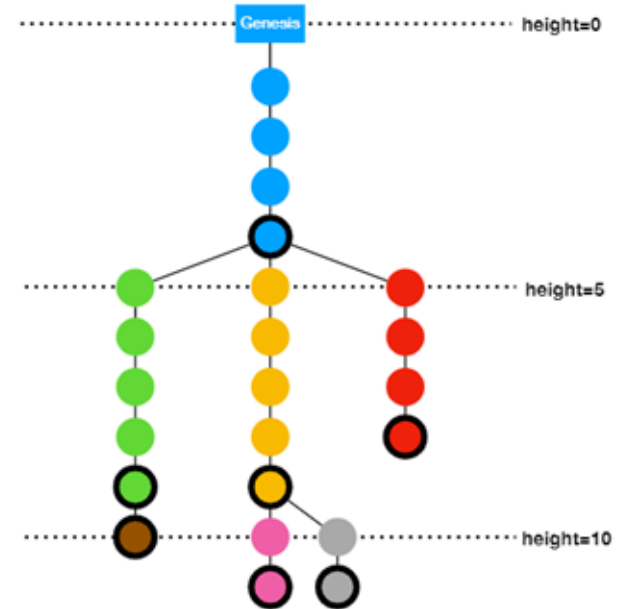


Analysis of private NaS

- Godfather-block: $\text{height}(b) \% c = 0$
- Only fork at the parents of godfather-blocks

c	1	2	3	4	5	6	7	8	9	10
ϕ_c	e	2.22547	2.01030	1.88255	1.79545	1.73110	1.68103	1.64060	1.60705	1.57860
β_c	$\frac{1}{1+c}$	0.31003	0.33219	0.34691	0.35772	0.36615	0.37299	0.37870	0.38358	0.38780

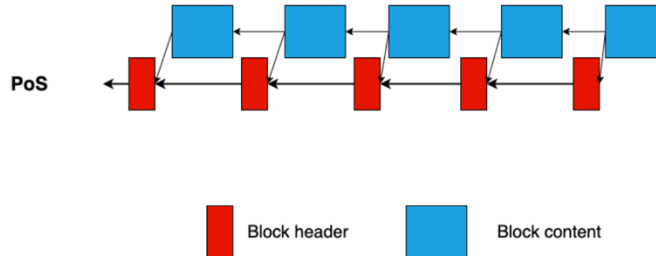
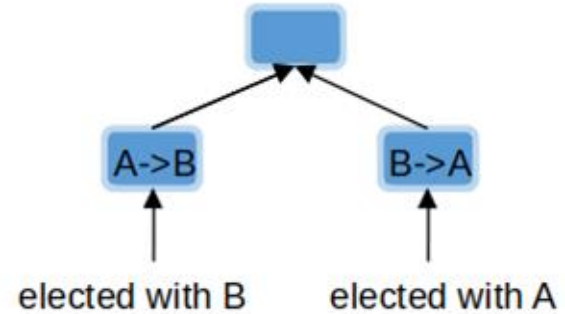
$\beta_c = 1/(1+\phi_c)$: threshold for private NaS attack



Proof of Stake

Dynamic stake

- Flaw of static stake
 - Prevent nodes from leaving and joining
 - A coin with no actual stake can participate
- What if stake is updated immediately?
 - Grinding attack: once the adversary is elected as a leader at round i , it can include transactions at round i to transfer all stake to a coin that has a higher chance of winning the election at round $i + 1$
- What if stake is updated with a delay of s blocks?
 - Long range attack: have a private chain with s blocks



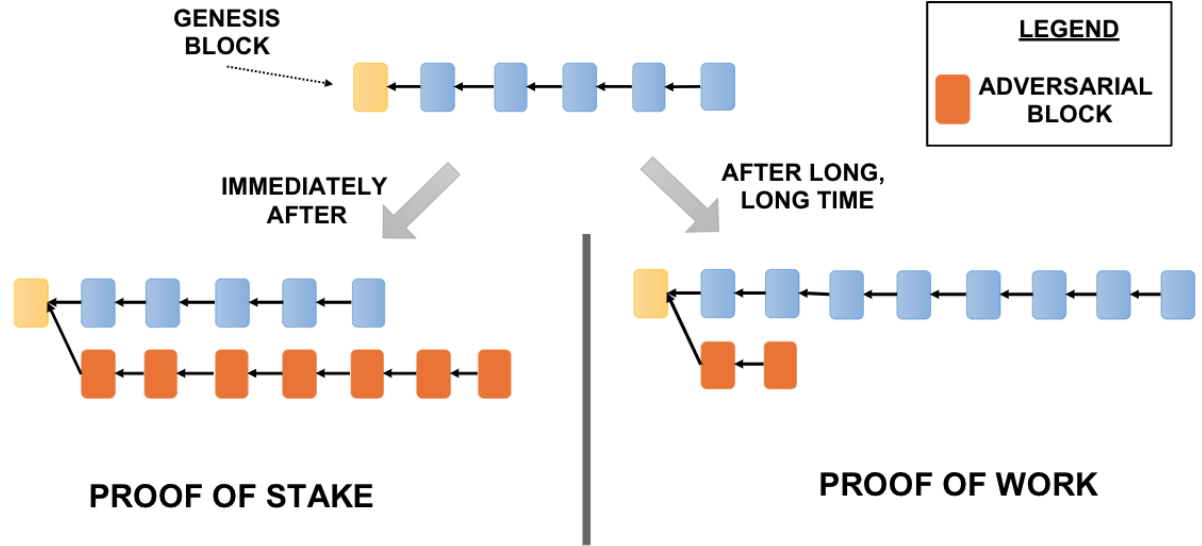
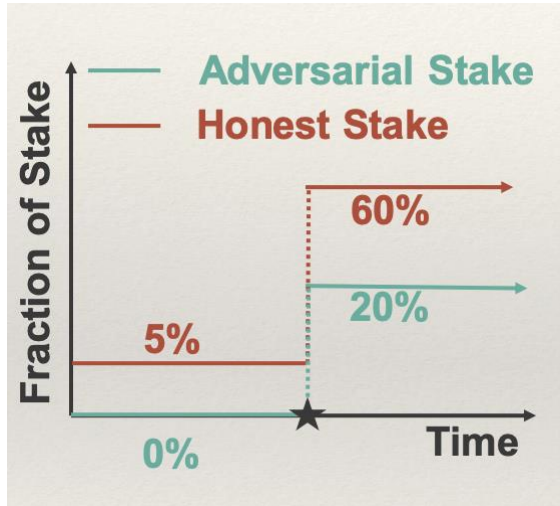
Long range attack

- In a long-range attack the victim tries to distinguish between two alternative histories furnished by the network without any recent information.
 - How does a new (or long term desynchronised) node synchronise with the blockchain?
- Mitigations
 - Trusted Checkpoints
 - New fork rule
 - VDFs

New Fork Choice Rule: s-truncation

- Stake is updated with a delay of s blocks
- Chain rule: When comparing two chains, both chains are truncated up to s blocks after the fork. Whichever truncated chain is mined in shorter time (and hence denser) is chosen to be mined on.

Dynamic availability



Crypto Primitive 3

Verifiable Delay Function (VDF)

$$\text{VDF}(sk , x) \rightarrow (y , \pi)$$

Takes L steps

$$\text{Verify}(pk , x , y , \pi) \rightarrow \text{True/False}$$

Takes much less than L steps

- Proof of sequential work

PoSAT: PoS with arrow-of-time

$$\text{VRF}(\text{RandSource}(\text{parent}), \text{ts}, \text{sk}_n) < T \cdot \text{stake}_n$$



$$\text{VDF}(\text{RandSource}(\text{parent}), \text{ts}, \text{sk}_n) < T \cdot \text{stake}_n$$

Resources

- ECE/COS 470, Pramod Viswanath, Princeton 2024